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The Willing but Wary Developer: AI Adoption, Trust Deficit, and Ethical Anxiety in Software Engineering

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AI Trust Research & Data analytics & Responsible AI Assistant

[Excel + Power BI + Microsoft Copilot / AI Chatbot]

A Collaborative Initiative

equipping Faculty with practical skills in AI, Data Analytics, and Business Intelligence

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The Willing but Wary Developer: AI Adoption, Trust Deficit, and Ethical Anxiety in Software Engineering

Project Synopsis

Project Title	The Willing but Wary Developer: AI Adoption, Trust Deficit, and Ethical Anxiety in Software Engineering
Domain	AI Trust Research& Data Analytics & Responsible AI
Tools Used	Excel + Power BI + Copilot Studio + AI Chatbot

1. Introduction

The software development profession is going through one of the most disorienting transformations it has seen in a long time. Artificial intelligence coding assistants, tools such as GitHub Copilot, ChatGPT, Google Gemini, and Amazon CodeWhisperer, have moved from experimental curiosities to everyday professional tools in less than three years. According to the **Stack Overflow Developer Survey 2025**, the world's largest annual study of the global developer community with 49,009 respondents across 177 countries, an extraordinary **84% of developers now use or plan to use AI tools in their daily work** ,up from 70% in 2023 (Stack Overflow, 2025). On the surface, this looks like a straightforward adoption success story. But looking more carefully at the same data reveals something genuinely unsettling: the very same developers who are embracing these tools are simultaneously losing faith in them. Trust in AI-generated code accuracy has fallen to just 33%, and only **3% of developers report high trust** in what these tools actually produce. Adoption is rising sharply. Trust is falling just as sharply. And both trends are accelerating at the same time.

This contradiction sits at the heart of what this project calls the **AI Trust Paradox** and it is not a single isolated problem. It is three business-critical challenges stacked on top of each other. First, developers continue using AI tools every day despite distrusting the code those tools generate, a pattern that creates real and measurable security risks in production environments (Pearce et al., 2022; Perry et al., 2023). Second, 81% of developers in the 2025 survey report serious security and privacy concerns about the very tools they depend on, yet most organisations have no formal AI governance framework in place a gap that leaves companies directly exposed to regulatory risk under the EU AI Act 2024 and India's DPDP Act 2023 (Morley et al., 2021; Holstein et al., 2019). Third, and perhaps most humanly significant, the share of developers who feel their job is safe from AI has declined

for three consecutive years, 70% in 2023, 68% in 2024, and 64% in 2025 and one in five developers now says that using AI tools has actively reduced their own confidence in their ability to solve problems independently. Developers are, in the most literal sense, becoming more productive and more anxious at exactly the same time (Walczok et al., 2025; Chuang et al., 2025).

The theoretical framework draws on three well-established models: the Technology Acceptance Model (Davis, 1989), which explains why adoption continues even when trust does not; Cognitive Dissonance Theory (Festinger, 1957; Marikyan et al., 2020), which explains the psychological mechanism that allows developers to rationalise continued use despite distrust; and the Job Demands-Resources Model (Bakker & Demerouti, 2017; Chuang et al., 2025), which frames AI tools as simultaneously a source of productivity gains and a source of anxiety.

2. Problem Statement

Software organisations worldwide are deploying AI coding tools at enterprise scale but they are doing so without any solid understanding of how their developers are actually experiencing those tools. The prevailing assumption is straightforward: if developers are using AI tools, they must trust them. The data tells a very different story. The Stack Overflow Developer Survey 2025 documents that adoption has reached an all-time high of 84% at the exact same time that trust has reached an all-time low of 33% a 51-percentage-point gap between the two metrics that has grown wider every single year from 2023 to 2025 (Stack Overflow, 2025). This is not a temporary adjustment effect that will resolve itself over time. It is a structural and worsening pattern with direct, measurable consequences for product quality, security posture, and the psychological health of the workforce. Classical technology acceptance frameworks, particularly Davis's (1989) Technology Acceptance Model, predict that adoption follows perceived usefulness and trust in output quality. When trust falls as adoption rises, the precise opposite of that prediction.

The most immediate and commercially costly dimension of this problem is code quality and security. Pearce et al., (2022) evaluated GitHub Copilot across 89 real-world security scenarios and found that approximately 40% of the programs it generated contained exploitable vulnerabilities. Perry et al., (2023) went further, demonstrating that developers who used AI assistants not only wrote less secure code than those who did not they were also more confident that their code was secure, a false assurance effect that makes the underlying risk substantially harder to catch before it reaches production. At the scale at which AI tools are now deployed across engineering teams globally, this compounds into dramatically higher security audit costs, growing technical debt, and a measurably expanded attack surface. IBM's (2024) Cost of a Data Breach Report puts the average breach cost at USD 4.88 million and AI-generated code that bypasses careful peer review makes that number more likely to materialise, not less.

The second dimension is what this project terms the **adoption-ethics gap** a disconnect between how widely AI tools are being used and how little governance infrastructure exists to manage that use responsibly. Across the 49,009 developers surveyed by Stack Overflow (2025), 81% reported active security and privacy concerns about the same AI tools they are already using daily. Holstein et al. (2019) surveyed 267 machine learning practitioners across major technology firms and found a consistent and significant gap between the ethical challenges that practitioners face on the ground and the governance support their organisations actually provide. Morley et al. (2021) confirmed that despite more than 80 AI ethics guidelines existing globally by that point, practitioners rarely apply them in practice — not out of indifference, but because the tools, training, and organisational structures needed to do so are largely absent from the workplace. For companies now operating under the EU AI Act 2024 or India's Digital Personal Data Protection Act 2023, this is no longer merely a reputational concern it is a live and growing legal compliance failure (Sanderson et al., 2023). Developers are using tools they do not fully trust, in environments without adequate governance, and most organisations are not measuring either of those facts.

The third dimension affects the human side of the business equation in ways that current HR and people management frameworks are not yet designed to address. The proportion of developers who believe AI does not pose a threat to their job has declined steadily and persistently: 70% in 2023, 68% in 2024, and 64% in 2025 a six-point decline over three consecutive years with no sign of stabilisation (Stack Overflow, 2025). More concerning still, one in five developers (20%) in the 2025 survey reports that using AI tools has actively reduced their own confidence in their ability to solve problems independently. This is not a marginal sentiment shift. Walczok et al. (2025) found that job insecurity actually increases alongside greater use of AI tools directly contradicting the widely held assumption that greater familiarity with AI reduces anxiety about it. Chuang et al. (2025), in a three-wave study of 600 participants, confirmed that AI-related technostress significantly increases exhaustion and reduces job satisfaction over time. For HR teams and talent managers, this creates a real and growing retention risk: experienced, high-performing developers who are going through professional identity erosion are precisely the people most likely to disengage or leave an organisation and they are the hardest and most expensive to replace.

3. Objectives

This project is guided by five focused objectives. Each is linked to a specific tool, a concrete deliverable, and a measurable outcome.

1. To clean and analyse the Stack Overflow Developer Survey 2025 dataset using Microsoft Excel, and to quantify all three AI Trust Paradox dimensions — use-despite-distrust, adoption-ethics gap, and utility-threat tension.

2. To design and publish a interactive Power BI dashboard that visualises the AI Trust Paradox findings across developer experience levels, countries, roles, and company sizes.

3. To deploy a Microsoft Copilot Studio chatbot connected to the Power BI dataset for natural language querying of findings, and to use an AI chatbot to produce structured, theoretically grounded interpretations of the results.

4. To develop a unified AI Trust Paradox Framework integrating the Technology Acceptance Model, Cognitive Dissonance Theory, and the Job Demands-Resources Model.

5. To document the complete project process, upload all artefacts, datasets, Excel workbook, Power BI dashboard , Copilot Studio configuration, chatbot prompt log, and this report — to a structured public GitHub repository, and to record a 10–15 minute demonstration video for submission.

5. Methodology / Implementation Plan

The following five-phase workflow provides a structured and reproducible approach to completing this project, integrating all four tools, Microsoft Excel, Power BI, Copilot Studio, and AI Chatbot across the phases.

#	Phase	Description
1	**Data Collection & Exploration**	The Stack Overflow Developer Survey 2025 dataset is publicly available on Kaggle (stackoverflow-developer-survey-2025) and at survey.stackoverflow.co/2025. It consists of two files: **survey_results_public.csv** (49,009 rows × 62 columns) and **survey_results_schema.csv** (variable definitions). Both files are loaded into Microsoft Excel. A **Variable Mapping Table** is constructed by cross-referencing the schema, assigning each key variable to one of the three paradox dimensions or to a demographic moderator.

2	**Data Cleaning & Preprocessing**	All data cleaning will be performed in Microsoft Excel. The trust variable AITrustOutput carries approximately 18% missing responses; these rows will be retained and recoded as a 'No Response' category rather than deleted, as prior research indicates that distrusting users are disproportionately likely to skip evaluative items — making the missingness pattern itself analytically meaningful. Multi-select fields including AIToolCurrently and AIFrustrations will be expanded from semicolon-delimited strings into binary dummy columns via Text-to-Columns.
3	**Analysis & AI Integration**	Core statistical analysis will be conducted in Microsoft Excel using Pivot Tables. For Paradox 1, daily AI tool usage will be cross-tabulated against trust level and disaggregated by experience group. Consistent with Stack Overflow's published data, developers with ten or more years of experience are expected to show both the lowest high-trust rate (approximately 2.6%) and the highest high-distrust rate (approximately 20%) — suggesting that professional expertise amplifies the paradox rather than resolving it. For Paradox 2, ethics concern rates will be calculated across adoption-breadth groups to examine whether concern rises with the number of AI tools used. For Paradox 3, the three-year job threat trend will be compiled from Stack Overflow's annual summaries and visualised as a line chart. Chi-square statistics and Cramér's V effect sizes will be calculated for all cross-tabulations. In parallel, ten structured AI chatbot prompts — one per key finding — will be used to generate theoretically grounded interpretations drawing on TAM, CDT, and JD-R theory, with every AI output validated against the dataset or cited literature before inclusion.
4	**Visualization / Output**	Import the cleaned dataset into Power BI Desktop and build a interactive report.
5	**Documentation & Submission**	Create a public GitHub repository with organised folders covering raw data, cleaned data, the Excel workbook, the Power BI .pbix file, Copilot Studio configuration, chatbot prompt logs, the report PDF, and the presentation slides.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core Ultra 5 and 16 GB RAM
- Operating System: Windows 11
- Microsoft Excel (Excel 2019)
- Power BI Desktop — latest version
- Microsoft Copilot — Microsoft 365 subscription
- AI Chatbot all
- Web browser — Google Chrome

B. Dataset

- Stack Overflow Developer Survey 2025 (Kaggle / Stack Overflow Open Data)

Dataset Name	Stack Overflow Developer Survey 2025
Primary Source	survey.stackoverflow.co/2025
Kaggle Link	https://www.kaggle.com/datasets/stackoverflow/stack-overflow-2025-developer-survey
Size	49,009 respondents | 177 countries | 62 variables
Licence	Open database licence (ODbL) — free for academic, research, and non-commercial use with attribution to Stack Overflow
Business Relevance	The dataset captures real developer behaviour at global scale across all major technology markets. Variables directly map to business outcomes: AI tool adoption rates, trust levels, ethics concerns, job satisfaction, and productivity impact making it uniquely suited for business-oriented AI governance analysis.
Key Variables	AIToolCurrently, AITrustOutput, AIFrustrations, AISentiment, AIThreat, AIProductivity, LLMsSelect, Country, YearsCode, DevType, OrgSize, EdLevel, JobSat

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset download, schema review, variable mapping table, and data inventory profiling in Excel	Week 1
2	Null handling, multi-select expansion, ordinal encoding, experience standardisation, and validation check	Week 2
3	AI chatbot prompt	Week 2 – 3
4	Power BI dashboard build	Week 3
5	Project report writing, GitHub repository, demonstration video recording Submission	Week 4

7. Expected Outcomes

This section summarises the seven concrete deliverables this project expects to produce by the end of the fellowship, each directly linked to one of the five objectives.

- **A fully cleaned, analysis-ready dataset with a documented quality log**
- Every data cleaning step will be recorded in a cleaning log that covers all issue types resolved, missing values, multi-select expansion, ordinal encoding, outlier standardisation, and validation checks
- **A interactive Power BI dashboard published to the cloud**

  The dashboard will include Trust Index KPI cards, the adoption-vs-trust trend line for 2023–2025, and dynamic filter slicers for Country, DevType, YearsCode Group, and Organisation Size .
- **A functional Copilot Studio AI chatbot with ten tested natural language queries**

  The chatbot will be connected live to the Power BI dataset and will correctly respond to queries covering each of the three paradox dimensions..
- **A validated AI chatbot prompt log with ten structured theoretical interpretations**

  Each of the ten prompts will generate a theoretically grounded interpretation of a specific statistical finding using TAM, CDT, or JD-R.

- **A unified three-dimensional AI Trust Paradox Framework**

  This framework will integrate TAM, Cognitive Dissonance Theory, and the Job Demands-Resources Model into a single, reusable theoretical model for explaining developer AI adoption behaviour. It will be the first framework of its kind for the global developer population, and will be accompanied by a complete academic synopsis targeting Computers in Human Behavior or IJIM.

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